INCH-POUND

MS21949B
20 September 2000
SUPERSEDING
MS21949A
13 May 1966

# DETAIL SPECIFICATION SHEET

## BOLT-CLUSTER FITTING, SINGLE PORT-FLARE

This specification is approved for use by all Departments and Agencies of the Department of Defense .

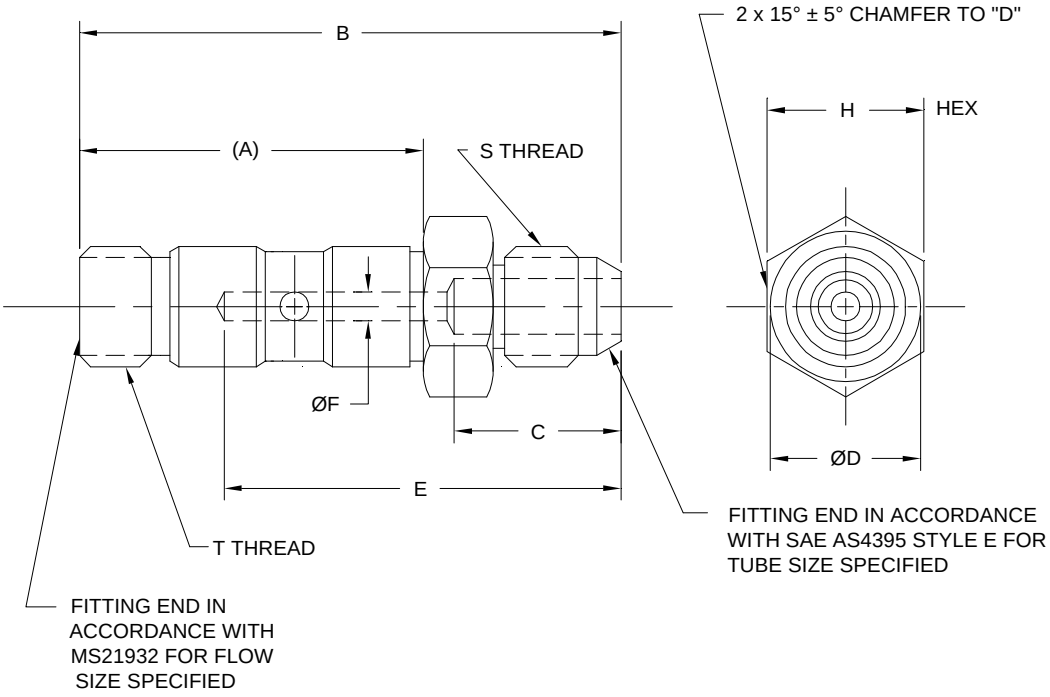


FIGURE 1. Bolt, Cluster Fitting.



AMSC N/A FSC 4730

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TABLE IA. Dimensions for low flow sizes

| Low flow size | | Tube OD | Thread S SAE AS8879 | Thread T SAE AS8879 |
|---|---|---|---|---|
| Al alloy 7075 | Al alloy 1/ 2014/2024 | | | |
| W3L | -3L | .187 | .375-24UNJF-3A | .4375-20UNJF-3A |
| W4L | -4L | .250 | .4375-20UNJF-3A | |
| W5L | -5L | .312 | .500-20UNJF-3A | |
| W6L | -6L | .375 | .5625-18UNJF-3A | |
| W8L | -8L | .500 | .750-16UNJF-3A | |

TABLE IA. Dimensions for low flow sizes - Continued

| A | B | C ± .062 | D ± .010 | E ± .062 | F ± .003 | H Hex + .003 - .010 |
|---|---|---|---|---|---|---|
| 1.438 | 2.229 | — | .605 | 1.625 | .125 | .625 |
| 1.438 | 2.332 | — | .668 | 1.750 | .172 | .688 |
| 1.438 | 2.394 | .750 | .730 | 1.812 | .203 | .750 |
| 1.438 | 2.431 | .750 | .855 | 1.812 | .203 | .875 |
| 1.438 | 2.532 | .750 | .980 | 1.937 | .203 | 1.000 |

1/ Aluminum alloy 2014 and 2024 parts are cancelled. Use Aluminum alloy 7075 parts for new design. Cancelled parts may be used till stock is exhausted.



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TABLE IB. Dimensions for high flow sizes

| High flow size | | Tube OD | Thread S SAE AS8879 | Thread T SAE AS8879 |
|---|---|---|---|---|
| Al alloy 7075 | Al alloy 1/ 2014/2024 | | | |
| W6H | -6H | .375 | .5625-18UNJF-3A | .5625-18UNJF-3A |
| W8H | -8H | .500 | .750-16UNJF-3A | |
| W10H | -10H | .625 | .875-14UNJF-3A | |

TABLE IB. Dimensions for high flow sizes - Continued

| A | B | C ± .062 | D ± .010 | E ± .062 | F ± .003 | H Hex |
|---|---|---|---|---|---|---|
| 1.688 | 2.681 | — | .855 | 2.000 | .297 | .875 + .003 - .010 |
| 1.688 | 2.782 | — | .980 | 2.125 | .391 | 1.000 + .003 - .010 |
| 1.688 | 2.883 | .937 | 1.103 | 2.188 | .422 | 1.125 + .003 - .012 |

1/ Aluminum alloy 2014 and 2024 parts are cancelled. Use Aluminum alloy 7075 parts for new design.

Cancelled parts may be used till stock is exhausted.

REQUIREMENTS:

1. The requirements for acquisition of the product described herein shall consist of this detail specification sheet and SAE AS4875/1.

2. Material: Aluminum alloy bar 7075-T73 per AMS-QQ-A-225/9.

3. Finish: See standard SAE AS4875. Aluminum alloy 7075 fittings shall be dyed brown.

4. All outside machined surfaces shall be finished to 125 µin $R_a$; all inside machined surfaces and hex flats of bar stock shall be finished to 250 µin $R_a$, unless otherwise specified. Surface finish shall be in accordance with ASME B46.1.

5. Fittings shall be free of all burrs and slivers.



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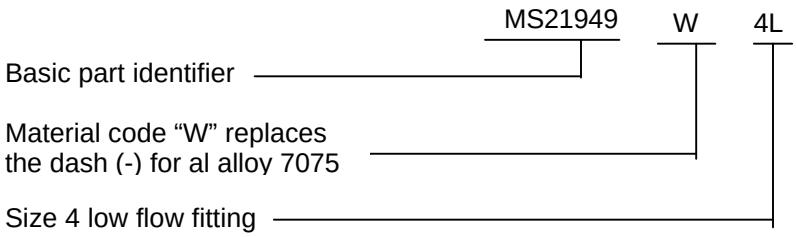
NOTES:
1. Dimensions in inches. Unless otherwise specified, tolerances: decimals ± .010, angles ± 1/2°.

2. Dimensioning and tolerancing per ASME Y14.5.

3. Part Identification number (PIN)

Example of PIN: MS21949W4L




Custodians:
Navy - AS
DLA - CC

Preparing activity:
DLA-CC

(Project 4730-2067)

Review activity:
Navy - SA



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